

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I K.Akshitha (192312527) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Akshitha

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

RESEARCH ARTICLE REVIEW

Decision Tree Based Machine Learning

Selected Article

Article Title: A Better Decision Tree: The Max-Cut Decision Tree with Modified PCA Improves Accuracy and Running Time

Authors: Jonathan Bodine and Dorit S. Hochbaum

Journal: SN Computer Science

Publisher: Springer Nature

Year: 2022

Volume: 3, Article 313

DOI: 10.1007/s42979-022-01147-4

Valid DOI: <https://doi.org/10.1007/s42979-022-01147-4>

The article is an Original Research article published in SN Computer Science in 2022 and is openly available through Springer.

Domain/Application Area: Machine learning, data mining, pattern classification, and high-dimensional data classification.

Specific ML Problem: The research addresses the limitations of conventional CART decision trees using Gini impurity, particularly their classification accuracy and computational efficiency when dealing with high-dimensional datasets.

TASK 1: ARTICLE SUMMARY

1(a) Citation Details and Research Problem

Bodine and Hochbaum proposed a modified decision-tree method called the Max Cut Node Means PCA decision tree. The method modifies the conventional CART decision tree in two important ways: it replaces the traditional Gini impurity splitting criterion with a Max Cut criterion, and it introduces Node Means PCA to obtain locally meaningful feature representations at every tree node.

The research focuses on classification problems where the number of features can be large and where conventional decision trees may have difficulty identifying useful splitting directions. The authors evaluated the proposed approach using more than 20,000 synthetic datasets and several real-world datasets, including Iris, Wine Quality, Indian Liver Patient Dataset, MNIST, CIFAR-10, and CIFAR-100.

The main objective is to improve both classification accuracy and computational speed while retaining the useful properties of decision trees such as relatively simple structure and interpretability.

1(b) Key Objective and Research Gap

The main objective of the research is to develop a decision-tree algorithm that performs better than conventional CART trees, especially for difficult and high-dimensional classification problems. Traditional CART trees normally select a feature and threshold using criteria such as Gini impurity. According to the authors, these criteria concentrate mainly on class separation and do not sufficiently consider the distances or similarities between observations in feature space.

The research gap identified by the authors is therefore the lack of a decision-tree construction method that simultaneously considers class separation, feature-space distance, local feature structure, accuracy,

and computational cost. Another limitation of conventional PCA preprocessing is that a single global transformation may not represent the characteristics of every sub-region of the dataset.

To address this gap, the authors proposed the Max Cut Node Means PCA approach. Max Cut considers both class membership and feature distance when selecting a split, while Node Means PCA calculates locally meaningful directions at every decision-tree node. The authors conducted extensive experiments on synthetic and real-world datasets to determine whether these modifications improve accuracy and runtime.

TASK 2: METHODOLOGY ANALYSIS

2(a) ML Techniques Used

1. Conventional CART Decision Tree

The baseline technique is the Classification and Regression Tree (CART) method. CART recursively selects a feature and threshold that produce the best separation of classes. The authors use Gini impurity as the baseline splitting criterion.

Gini impurity can be represented as:

$$\text{Gini} = \sum p(c)(1 - p(c))$$

where (p_c) represents the proportion of samples belonging to class (c).

2. Max Cut Splitting Criterion

The first major modification is the Max Cut splitting criterion.

Instead of considering only class impurity, Max Cut considers the distance between observations belonging to different classes on opposite sides of a candidate threshold.

In simple terms:

Gini:

Select a split mainly according to class purity.

Max Cut:

Select a split that separates different classes while also maximizing separation between observations in feature space.

The authors state that the proposed Max Cut split can be implemented in ($O(n \log n)$) time for a given feature, which is asymptotically comparable to the conventional splitting procedure.

3. Node Means PCA

The second major modification is Node Means PCA.

Instead of performing PCA once on the entire dataset, PCA is performed locally during tree construction. At each node, the method calculates class-related mean representations and uses PCA to identify meaningful directions for that particular subproblem.

This provides two important benefits:

It creates a feature representation specific to the current decision-tree node.

It can reduce dimensionality and therefore reduce computational requirements.

The authors also examined another method called Node Features PCA, where PCA is performed directly on the observations reaching each node.

Dataset Used

The authors used both synthetically generated datasets and real-world datasets.

Synthetic Dataset

The synthetic datasets were generated using the Scikit-learn `make_classification` function. The authors controlled:

- 1.Number of training examples
- 2.Number of classes
- 3.Number of informative features
- 4.Number of noisy features

The test set was generally fixed at 100,000 observations. The synthetic data contained class clusters generated around hypercube vertices, with normally distributed observations and approximately 1% randomly assigned class labels.

More than 20,000 synthetic datasets were evaluated, with training sizes ranging from approximately 100 to 300,000 observations. Both binary and ten-class classification problems were considered.

The ILPD dataset originally contains 583 samples, but the authors removed samples containing missing values, leaving 579 observations.

Data Preparation

For synthetic experiments, the datasets were standardized using Scikit-learn `StandardScaler`, with the scaler fitted on the training data. This was important because the proposed Max Cut criterion and PCA use feature distances and directions.

For real-world datasets, the researchers tested variants both with and without feature standardization and reported the better result. Small datasets were evaluated using 10×10 cross-validation, while MNIST, CIFAR-10, and CIFAR-100 used an 80%–20% train-test split because repeated cross-validation would require substantial computation.

2(b) Mapping to CO4 Topics

CO4 Topic	Connection with Article
Inductive Learning	The decision tree learns classification rules from training examples generalizes them to unseen observations.
Decision Trees	The complete research focuses on improving CART decision construction.
Statistical Learning	Statistical significance was evaluated using paired sample t-tests Holm-Bonferroni correction.
Reinforcement Learning	Not directly used in this research.

Methodological Strength

A major strength is the extensive experimental evaluation. The authors did not depend on only one dataset. They evaluated more than 20,000 synthetic datasets and several real-world datasets with

different numbers of classes, features, and noise levels. Statistical significance testing was also performed using paired t-tests with Holm-Bonferroni correction.

Methodological Limitation

One limitation is that a substantial portion of the evaluation depends on synthetically generated datasets. Synthetic data provides excellent experimental control, but its statistical structure may not represent all real-world applications. In addition, the real-world evaluation uses only a limited collection of benchmark datasets. Therefore, further testing on large, domain-specific datasets would be necessary before claiming universal superiority.

TASK 3: RESULTS AND CRITICAL EVALUATION

3(a) Key Results

The article reports substantial improvements in both classification accuracy and computational efficiency.

The most important result is observed on CIFAR-100. The Max Cut Node Means PCA approach achieved an accuracy of 0.124 compared with 0.083 for the baseline Gini CART, representing a reported 49.4% improvement in accuracy performance.

The method also achieved major computational improvements. For CIFAR-100, the authors report approximately 94% reduction in CPU time compared with the baseline variant. The optimized MaxCutTree implementation achieved approximately $6.8\times$ speed-up compared with Scikit-learn CART.

For the synthetic experiments, Max Cut Node Means PCA also showed strong results. In one comparison involving 5,640 datasets, its mean accuracy ratio relative to Gini Node Means PCA was 1.0003, with a 95% confidence interval of 1.00008–1.00053.

3(b) Critical Evaluation

Are the Reported Metrics Realistic?

The results are reasonably convincing because the authors use a large number of independent synthetic datasets and statistically test their comparisons. The use of real-world datasets also provides additional evidence that the proposed method is not limited entirely to artificial data.

However, the reported improvements should be interpreted carefully.

For example, the CIFAR-100 accuracy improves from 8.3% to 12.4%. Although this is a relative improvement of 49.4%, the absolute accuracy remains low. Therefore, the percentage improvement can appear much more impressive than the actual classification capability.

This is an important point when evaluating machine-learning research: relative improvement and practical performance are not always the same thing.

Potential Dataset Bias

The synthetic datasets were generated according to controlled assumptions, including cluster structures and normally distributed observations. Real-world data can have missing values, class imbalance, temporal dependencies, outliers, and complicated feature relationships.

The real-world datasets are also primarily established benchmark datasets. Therefore, additional evaluation on modern industrial datasets would improve confidence in generalization.

Evaluation Protocol

The evaluation protocol is generally strong because:

- Multiple datasets were tested.
- Both binary and multiclass classification were evaluated.
- Training size and feature dimensionality were varied.
- Noise was deliberately introduced.
- Statistical significance testing was performed.
- Runtime was measured.

However, different evaluation protocols were used for small and large datasets: 10×10 cross-validation for several smaller datasets and an 80%–20% split for MNIST/CIFAR datasets.

Additional Experiments Recommended

1. The research could be strengthened by:
2. Testing on modern industrial datasets.
3. Performing repeated experiments with different random seeds.
4. Reporting confidence intervals for every real-world dataset.

The authors themselves identify ensemble methods as an important direction for future research.

3(c) Real-World Applicability

The proposed method can potentially be applied wherever classification involves a large number of features.

Possible Applications

- Medical diagnosis
- Image classification
- Industrial fault detection
- Cybersecurity intrusion detection

The method is especially attractive for high-dimensional datasets, because the research demonstrates that its accuracy and computational benefits become more significant as dimensionality increases.

Challenges in Industry

1. Data availability:

Companies may not have sufficiently large and clean labelled datasets.

2. Data quality:

Real-world data contains missing values, outliers, noise and inconsistent measurements.

3. Computational requirements:

Although the proposed method improves runtime, PCA and tree construction can still become computationally expensive for extremely large datasets.

4. Model maintenance:

Industrial datasets change over time. A decision tree may need retraining when the underlying data distribution changes.

TASK 4: REFLECTION AND LEARNING OUTCOME

4(a) Three Key Insights Gained

Insight 1: Decision-tree performance depends strongly on the splitting criterion

Before studying this article, a decision tree can appear to be simply a structure of nodes and branches. The article demonstrates that the mathematical criterion used to select each split can significantly affect classification performance.

This connects directly to the Decision Trees topic in CO4. Traditional CART commonly uses Gini impurity, whereas this research introduces Max Cut as an alternative criterion.

Insight 2: Feature representation can be adapted during inductive learning

A major learning outcome is that preprocessing does not necessarily have to occur only once before model training. The authors perform PCA locally at individual tree nodes.

This connects to Inductive Learning because the model continuously learns useful decision boundaries from subsets of the training data. Node Means PCA allows the representation of the data to change according to the classification problem at each node.

Insight 3: Statistical significance is important when comparing ML algorithms

The article does not simply claim that one method is better because its average accuracy is higher. The authors evaluate thousands of datasets and use paired statistical tests with Holm-Bonferroni correction.

This connects to Statistical Learning Methods in CO4. It demonstrates that a small difference in accuracy should not automatically be interpreted as a meaningful improvement unless statistical evidence supports it.

4(b) Proposed Extension / Novel Experiment

Proposed Experiment: MaxCutTree for Cybersecurity Intrusion Detection

I would extend this research by applying the Max Cut Node Means PCA decision tree to a modern network-intrusion dataset, such as a contemporary cybersecurity traffic dataset containing normal and malicious network flows.

The experiment would compare:

- Standard CART
- Random Forest
- XGBoost
- Support Vector Machine
- Max Cut Decision Tree

The experiment would evaluate:

- Accuracy
- Precision
- Recall
- F1-score

This extension is feasible because cybersecurity datasets generally contain many network-flow features, making them suitable for evaluating the high-dimensional classification capability of the proposed method.

The expected impact is significant because intrusion detection requires both high detection performance and fast classification. If MaxCutTree can maintain good F1-score and recall while reducing computational cost, it could be useful for real-time network security systems.

OVERALL CONCLUSION

The selected research article provides a strong contribution to the Decision Tree and Inductive Learning areas of machine learning. Bodine and Hochbaum identify limitations in traditional CART decision trees and propose two major modifications: the Max Cut splitting criterion and Node Means PCA. The combination forms the Max Cut Node Means PCA decision tree.

The experimental evaluation is extensive, involving more than 20,000 synthetic datasets and several well-known real-world datasets. The results show that the proposed method can improve accuracy and reduce computational time, particularly for high-dimensional classification problems. On CIFAR-100, for example, the method increased reported accuracy from 8.3% to 12.4%, corresponding to a 49.4% relative improvement, while the optimized implementation achieved approximately 6.8× speed-up compared with Scikit-learn CART.

From a CO4 perspective, the article demonstrates how inductive learning, decision-tree splitting, feature representation, and statistical evaluation work together in an ML system. It also shows that improving an algorithm is not only about increasing accuracy; computational efficiency and robustness to high-dimensional data are equally important.

Overall, the Max Cut Node Means PCA approach is a valuable research contribution because it attempts to make traditional decision trees more effective for modern, high-dimensional classification problems. Further research involving modern datasets, ensemble methods, cybersecurity, healthcare, and industrial applications would help establish its practical value.